Janith Wanniarachchi

PhD candidate, Monash University: explainable AI and data visualisation
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Profile

Statistician and R developer. I research how to make machine learning explanations visible and trustworthy, and I build open-source software to do it. I have industry experience in Shiny and R package engineering, I teach at Monash, and I have a track record of community talks and workshops.

Education

2023 – now **PhD, Econometrics and Business Statistics**, Monash University, Australia
Thesis: *Explaining AI through visual explanations and efficient interpretable models.* Supervisors: Dianne Cook, Patricia Menendez, Thiyanga Talagala, Kate Saunders. Australian Government Research Training Program (RTP) Scholarship.

2017 – 2022 **BSc (Honours) in Statistics, First Class**, University of Sri Jayewardenepura, Sri Lanka
Minors in Computer Science and Mathematics. Statistics GPA 3.88; overall GPA 3.82.

2013 **Diploma in Web Engineering**, ESOFT Metro Campus

Research projects

Chapter 2 **Visual explanations of XAI explainers**: geometric representations of LIME, SHAP, counterfactuals and Anchors in the data space.

Chapter 3 **Random seeds and the complexity–parsimony trade-off**: the seed sensitivity of simple neural networks, with a Shiny teaching tool.

Chapters 4–6 **kultarr, kumquat and sillysplines**: transparent R implementations of Anchors and LIME, and an interactive generator for synthetic data.

Publications & presentations

2026 **Wanniarachchi, J.** *Visual Explanations of XAI Explainers.* useR! 2026.

2023 **Wanniarachchi, J.** *Seeing the Smoke Before the Fire: Visualising Spatiotemporal Indicators of Bushfires through Counterfactual Explanations of Machine Learning Models.* 12th IASC-ARS Conference, Macquarie University, Sydney.

2022 **Wanniarachchi, J. C.**, & Talagala, T. S. *scatteR: Generating instance space based on scagnostics.* arXiv:2209.06682. Presented at useR! 2022.

2022 Karunanayake, V. J., **Wanniarachchi, J. C.**, Karunanayake, P. N. K., & Rajapaksha, U. U. S. *Intelligent System to Verify the Effectiveness of Proposed Teacher Transfers Incorporating Human Factors.* ICARC 2022, IEEE. doi:10.1109/ICARC54489.2022.9754089

Software

Author **kumquat** (CRAN): transparent LIME · **scatteR** (CRAN): scagnostics-driven data generation · **kultarr**: Anchors in pure R · **sillysplines**: interactive spline data generator

Contributor **DSjobtracker**, **nic** and **tsdataleaks** (all on CRAN): data curation, colour palettes, and Rcpp performance work

Experience

2023 – 2024 **Teaching Associate**, Monash University
Tutorials for ETC5523 *Communicating with Data* (Shiny, R packages, storytelling with visualisation) and ETC5512 *Wild-Caught Data* (collecting and wrangling tabular and geospatial data).

2022 – 2023 **R/Shiny Developer**, Appsilon
Built Shiny dashboards and provided software engineering support for R package development for life-sciences clients.

2021 – 2022 **Contractor Consultant**, labml.ai
Analysed online user segments and feature usage, and recorded research paper explainers (e.g. DeepMind's RETRO).

2020 – 2021 **Trainee Data Analyst**, Trabeya
Built the deduplication component of an automated data-cleaning product and a session-based recommender served through a real-time microservices backend.

2021 **Consultant Data Analyst**, CreativeHub
Built a hypothesis-testing framework and dashboards for Facebook ad campaigns.

2019 **Contractor Consultant**, Intellimorph
Built anomaly detection for robotic process automation and a computer vision system for a robotic hand.

2021 **Consultant**, Statistical Consultancy Service, University of Sri Jayewardenepura
Curated and cleaned the 2021 DSjobtracker data for CRAN.

Awards

2023 **United Nations Datathon**: 2nd place globally and 1st in the Oceania region (team of five; solar-farm siting web app).

2023 **Outstanding Young Researcher / Student Contributed Oral Presentation Award**, IASC-ARS 2023.

2022 **SLAAS Physical Science Award in Statistics**, Sri Lanka Association for the Advancement of Science, for the best BSc Statistics (Hons) research.

2022 **useR! 2022 diversity scholarship**; **top 10 finalist**, Octave Advanced Analytics Symposium; nominee for the Most Venerable Sri Sumangala Gold Medal, University of Sri Jayewardenepura.

2017 – 2020 1st runner-up at hackStat 2.0 (2019) and the SLIIT Codefest Datathon (2019); finalist at Data Storm 2.0 (2020) and the Dialog Datathon; honourable mention at the Asia-Pacific Informatics Olympiad (2017); IEEEXtreme 12.0 global rank 487.

Selected talks

2019 – 2026 useR! 2026 · IASC-ARS 2023 · useR! 2022 · Abuja useR Group (2022) · EDEX Expo (2022) · R-Ladies Colombo (2021) · Fundamentals of Data Visualisation and Tableau workshops (2021) · GDG ML Bootcamp (2019) · TensorFlow All Around (2019). Full list at janithwanni.netlify.app/talks.

Leadership & service

2022 **Session chair**, useR! 2022 (Inference Methods).

2020 – 2021 **President**, Statistics Society, University of Sri Jayewardenepura. Founded the Pearl Viz data visualisation competition and ran the "Ask the Experts" alumni panel (200+ students) and the Outliers student writing programme.

2019 **Head**, TensorFlow User Group Sri Lanka. Organised TensorFlow All Around 2019.

Skills

Languages R (Shiny, package development, S7, Rcpp, tidyverse), Python (PyTorch, TensorFlow), JavaScript/Svelte, SQL, C++

Tools Quarto, Git, Azure SQL, Google Cloud, Tableau, Power BI

English IELTS overall 8.0 (Listening 9.0, Reading 9.0)